E N G I N E E R I N G R E A L I T Y C H E C K

MAYA Build Constraints and Risk Register

Companion to PRD v0.1 - problems found, problems ahead, and what cannot be done

Product: MAYA - AI That Protects (voice anti-spoofing)

Context: hridhAI Hackathon 2026, 48-hour MVP build

Basis: PRD v0.1 and the pre-build design review, 6 November 2026

Audience: build team - internal, blunt by design

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1. Executive Summary

This document is the engineering companion to MAYA PRD v0.1. It records every feasibility problem surfaced during the pre-build design review: problems that exist today, problems the team will hit during the 48-hour build, and capabilities that are not achievable on the target platform at any level of effort. Its purpose is to force wording and scope corrections before hour 0, so that no build hour is spent discovering a platform wall at hour 40.

Three findings dominate. First, the PRD's central technical premise - capturing the caller's voice from a live call via AudioRecord - does not work on stock Android; downlink capture is a privileged path that returns silence on most devices, so the capture design must be reworked around speakerphone-plus-microphone capture with automatic speaker enablement. Second, several PRD claims are impossible as stated (we stop voice cloning; first result under 4 seconds at full confidence; the RECORD_AUDIO phrasing) and must be replaced with the defensible wording set out in Section 4 and Appendix A. Third, what remains is genuinely buildable: a quantized pretrained AASIST-class model, a native Kotlin pipeline and a staged verdict UI fit inside the 200-300 MB RAM budget and the 48-hour schedule, provided the hour-12 fallback rules in Section 5 are enforced.

Every entry carries a status. BLOCKED means the platform forbids it for any third-party app. REWORK means the PRD design as written will not work but a concrete alternative exists. RISK means a probable build-phase or demo failure with a defined mitigation. VERIFY means a fact assumed during review could not be confirmed and must be checked (Section 8). DECIDE means an open decision with a recommendation attached (Section 9). Owners for verification items are assigned at the sign-off meeting before hour 0; until then, nothing in Section 8 may be treated as established fact.

2. Scope, Sources and Status Codes

Scope. The register covers three time horizons: constraints that are true today and permanent on stock Android (Section 3); product claims that are impossible regardless of implementation quality (Section 4); and probabilistic risks expected during the 48-hour prototype build and on demo day (Sections 5 to 7). Sections 8 to 10 record what must be verified, what must be decided, and how specific PRD v0.1 sentences change as a result.

Sources. MAYA PRD v0.1 (draft for hridhAI Hackathon 2026); the pre-build design review sessions of 6 November 2026; Android platform behavior as known to the team; and anti-spoofing literature where cited inline. Where a vendor-specific or paper-specific figure could not be re-verified during review (the research lookup tool was unavailable that day), the item

is carried as an assumption and listed in Section 8 rather than stated as fact. This is deliberate: the register prefers an honest unknown over a confident invention.

Table 2-1: Status codes used throughout this register

Code	Meaning	Team action
BLOCKED	Not achievable on stock Android by any third-party app; enforced by the platform, not by skill.	Do not spend build hours on it; use the listed workaround or cut the feature.
REWORK	The PRD design as written will not work; a concrete alternative exists.	Rewrite the affected PRD section before hour 0.
RISK	Probable failure during the 48-hour build or on demo day; mitigation is defined.	Assign the mitigation and respect the decision trigger.
VERIFY	Fact assumed but not confirmed during review.	Verify per Section 8 before it becomes load-bearing.
DECIDE	Open product or architecture decision.	Sign off per Section 9 by the stated deadline.

3. Hard Platform Walls: Not Doable on Stock Android

The constraints below are enforced by the Android platform, not by team skill or effort. Each has either a workaround that changes the product story or a scope cut. None of them can be engineered away within a third-party app without system privileges, root access or an OEM partnership. The single most consequential item is W1: it shifts the product story from a transparent always-on shield to an auto-speaker shield with an explicit cannot-scan state for earpiece-only calls. That shift belongs in the pitch before a judge finds it first.

Table 3-1: Platform-enforced constraints and workarounds

#	Constraint	Why it is not doable	Impact on PRD v0.1	Workaround and status
W1	Capturing the caller's voice (downlink) directly	VOICE_DOWNLINK and VOICE_CALL AudioRecord sources are privileged; most OEMs return silence, and call-recorder apps broke system-wide after Android 10.	Section 6 P0-2 capture design invalid as written; the always-on-without-speaker goal is unreachable.	Capture acoustically: speakerphone plus MIC or UNPROCESSED source; auto-enable the speaker on call connect; build the AudioSource abstraction (call-mic, file, loopback) so the pipeline is testable without live calls. REWORK.
W2	Perturbing our own outgoing voice on carrier calls	No app-level injection point exists into the modem's uplink audio path - symmetric to W1.	Stopping cloning of the speaker's live voice applies only to calls made through MAYA.	MAYA Live VoIP mode (roadmap V3) where MAYA owns the full audio stack; on PSTN calls, detection (Guard) remains the only line of defense. BLOCKED for carrier calls.
W3	Programmatically ending the call	TelecomManager.endCall() effectively requires default-dialer or system privileges; AccessibilityService clicks on the end-call button are fragile across OEM in-call UIs.	The RED verdict HANG UP action degrades to a one-tap assisted hang-up.	A single red END CALL button: try the privileged path when available, fall back to the accessibility click; the user always confirms. REWORK.
W4	NNAPI acceleration on current Android	NNAPI is deprecated as of Android 15; Google's stated direction is LiteRT with CPU and vendor delegates.	Section 7 names the NNAPI Delegate as the acceleration path.	Default to LiteRT (TFLite successor) with the XNNPACK CPU delegate; treat the GPU delegate as optional. Confirm on the demo device (U7). REWORK.

#	Constraint	Why it is not doable	Impact on PRD v0.1	Workaround and status
W5	Reliable always-on background operation	OEM ROMs common in India (Xiaomi, Realme and similar) aggressively kill background apps; API 34+ requires explicit foregroundServiceType microphone and phoneCall with matching permissions, or in-call mic access fails silently.	The always-on and battery claims are at risk on real devices.	Correct service types and permissions; battery-optimization exemption requested on first run; test on one Xiaomi-class device. RISK.
W6	Refreshing the offline scam-number blocklist	The APK has no INTERNET permission by design; there is no runtime network path.	Proactive pre-ring protection ages between app releases.	Ship a starter blocklist inside the APK; refresh only via app releases, which keeps the privacy invariant intact. A restricted update channel is decision D2. DECIDE.
W7	An offline proof performed as a live call in airplane mode	Airplane mode terminates cellular connectivity; a live PSTN call cannot exist in the very state being demonstrated.	Demo script step 1 as written is self-contradictory.	Prove offline-ness structurally (APK manifest with no INTERNET permission, shown on stage) plus a FileSource scan of bundled audio running the identical pipeline. REWORK.

4. Claims We Cannot Make (Threat-Model Limits)

The items below are threat-model impossibilities, not engineering gaps. Each row pairs the impossible claim with the defensible replacement. Using an original claim in front of an informed judge invites a one-question kill - your child's videos are already public, so what exactly did you stop? The replacement claims are written to survive that question, and they are stronger for it: precision about limits reads as competence, not weakness.

Table 4-1: Impossible claims and their defensible replacements

#	Claim as drafted	Why it is impossible	What we can honestly claim
C1	We stop voice cloning.	Cloning happens offline on attacker hardware from audio that is already public (social posts, forwarded clips). No receiver-side app can retroactively protect that audio. Even publish-side perturbation (Shield) only degrades future harvesting and is sensitive to codecs and adaptive attackers.	Guard detects cloned voices during the call; Shield makes future published and in-MAYA-call voice materially harder to harvest. Nothing retroactively protects audio that is already public - which is exactly why detection is the last line.
C2	Voice analysis identifies scammers.	Most live scam callers are humans; their voice is authentic, so the authenticity engine returns GREEN while the scam proceeds.	Dual engine: voice authenticity at the signal level plus on-device scam-script detection at the transcript level (needs 10-20 seconds of speech). The two verdicts are displayed separately because they call for different responses.
C3	First result in under 4 seconds.	The stated decision rule (3-second window, vote over the last 3) completes at roughly 9-11 seconds; a 4-second result is necessarily single-window and provisional.	Provisional verdict at about 4 seconds; confirmed three-window verdict at about 9 seconds; scam-script alerts at 15-20 seconds as evidence accumulates.
C4	No RECORD_AUDIO permission / no audio persistence as phrased	Microphone capture technically and legally requires RECORD_AUDIO; the promise as phrased is unverifiable and easily attacked.	RECORD_AUDIO is requested for capture only: audio lives in an in-memory buffer wiped after inference, is never written to storage, and the APK declares no INTERNET permission at all.
C5	Offline scam-number protection, always current.	No INTERNET permission means no runtime list refresh (W6).	An offline starter blocklist of reported fraud patterns, refreshed with each app release.
C6	Our anti-cloning poisoning defeats ElevenLabs-class cloning.	Adversarial perturbations partially strip out through lossy codecs (AMR/Opus, messaging-app re-encoding), weaken under streaming chunking, and transfer to closed production encoders is not guaranteed. The technique family is research-grade.	Raises the attacker's cost: harvested audio is degraded in quality. The demo shows the side-by-side degradation delta, not a defeat claim.

#	Claim as drafted	Why it is impossible	What we can honestly claim
C7	9 out of 10 cloned samples flagged correctly, live.	ASVspoof-2019-trained models are documented to degrade on in-the-wild and modern neural-TTS audio, and telephony band-limiting removes many high-frequency artifacts. An accuracy number is only defensible on the team's own channel-simulated calibration corpus.	9 out of 10 on our calibrated, channel-degraded test corpus; the stage demo uses pre-tested material and calibrated thresholds.

5. Build-Phase Risk Register (48-Hour Prototype)

Probabilistic failures expected during the build, each with a mitigation and the point in the schedule where it must be resolved. Severity: H means demo-killing if unresolved, M means the result is degraded, L means cosmetic or recoverable. Two entries deserve emphasis. R2 is the trap the team is most likely to dismiss: the PRD's named audio source applies acoustic echo cancellation, which treats the caller's loudspeaker voice as echo and suppresses it. R3 is the trap most likely to survive until the demo: a model calibrated on clean broadband audio will not behave the same through a phone channel, so the calibration corpus must be channel-degraded on purpose.

Table 5-1: Risks, mitigations and decision triggers

ID	Risk	When	Sev	Mitigation and decision trigger
R1	AASIST graph-attention layers fail ONNX/TFLite export, or converted scores drift from PyTorch.	Hours 0-12	H	Dry-run the export in hour 0. If still blocked by hour 12, swap to a naturally exportable architecture (RawNet2, LCNN-lite). After any conversion, require PyTorch-versus-TFLite score parity within about 1e-3 on ten fixed clips before trusting the device.
R2	Acoustic echo cancellation in VOICE_COMMUNICATION suppresses the far-end voice during speakerphone capture.	Hours 6-18	H	Run the capture-source matrix in the first 6 hours: MIC, UNPROCESSED and VOICE_COMMUNICATION, speaker on and off, on two devices. Pick the source empirically; do not assume.

ID	Risk	When	Sev	Mitigation and decision trigger
R3	Generalization gap: model trained on clean broadband audio versus telephony-band, modern-vocoder input.	Hours 12-30	H	Build the calibration corpus before the app: cloned plus genuine samples pushed through a 300-3400 Hz bandpass, an AMR encode-decode roundtrip and the speaker-to-mic path. Fit thresholds on degraded audio only. No model training in 48 hours - calibration and thresholds only.
R4	False RED verdict on a genuine caller.	Hours 18-48	H	Asymmetric thresholds: RED requires stronger evidence plus three-window confirmation; an amber provisional state absorbs early uncertainty. A false accusation is the product's worst failure mode.
R5	Ringtones, DTMF tones, hold music and silence scored as speech.	Hours 18-30	M	Energy or VAD gate before inference; non-speech windows are skipped, not scored.
R6	RAM overrun once the scam-intent engine is added.	Hours 18-36	M	Tiered activation: streaming ASR loads only while a call is live and frees on disconnect; keyword spotting is the first rung of the ladder; native Kotlin shell instead of Flutter (D3, D4).
R7	Battery drain and Doze throttling during long calls.	Post-build	M	Foreground service with microphone and phoneCall types (W5); measure drain for one call-hour once, record the figure, and put the measurement in the deck.
R8	AccessibilityService invites Play policy friction.	Post-hackathon	L	Cut from the MVP unless a specific need is proven; TelecomManager callbacks plus CallScreeningService cover the requirements.
R9	Stage logistics: cellular dependency and judge-voice cloning turnaround.	Demo day	H	Two-phone rig; clones rendered the night before; judge recording done 30 minutes prior with a teammate fallback take ready; FileSource scan button as the on-stage plan B; the final 6 hours are reserved for rehearsal only.
R10	Unverified facts silently shaping the design.	Any time	M	Section 8 list assigned before hour 6; after that point, no VERIFY item may remain load-bearing.

6. Resource Budget Reality (RAM, Latency, Battery)

The RAM targets stated in review - 200 MB preferred, 300 MB ceiling - are achievable, but only with a tiered architecture and a native shell. All figures below are planning estimates built from component-typical values; they must be confirmed by on-device measurement

before any number is externalized (items U3 and U7). The structural conclusion does not depend on the exact values: the voice engine fits comfortably, and the scam engine is what forces tiering.

Table 6-1: RAM planning budget by component

Component	Planning estimate	Notes
Native Kotlin app plus foreground service	40-60 MB	Baseline for the shell, overlay and call-state plumbing.
Flutter engine (if retained)	+20-40 MB	Dead weight for a background-service product; decision D3 recommends cutting it.
Voice-authenticity engine (AASIST-class int8/FP16, arena, delegate)	20-40 MB	Weights are only about 0.8-3 MB; the arena and delegate dominate.
Keyword-spotting scam detector	5-10 MB	First rung of the scam-engine ladder.
Streaming ASR (Vosk-class or whisper-tiny int8)	60-120 MB	Second rung; loaded only while a call is live, freed on disconnect.
Audio buffers and DSP features	under 5 MB	3 seconds at 16 kHz mono is about 96 KB per window; circular buffer.

Verdict on RAM. Voice-only operation lands around 90-140 MB - comfortably inside 200 MB. The dual-engine configuration with streaming ASR lands around 180-280 MB - inside the 300 MB ceiling but tight against 200 MB. If the 200 MB figure is hard, the keyword-spotting rung keeps the product inside budget with reduced scam coverage; the full-ASR tier then becomes a device-capability feature rather than a promise.

Table 6-2: Verdict timing math (3 s window, 2 s hop, vote over 3 windows)

Stage	When	Basis
Provisional verdict	t = 3.5-4 s	First 3-second window captured plus single-window inference under 800 ms; shown in amber unless the score is extreme.
Confirmed verdict	t = 9-11 s	Three windows at a 2-second hop, majority vote as specified in the PRD.
Scam-script alert	t = 15-20 s	Transcript evidence needs sustained speech; keyword-spotting hits can alert earlier.
Non-speech windows	skipped	VAD gate: no verdict updates from silence, ringback tones, DTMF or hold music (R5).

Battery. The duty cycle - 3-second capture every 2 seconds, sub-second LiteRT inference with XNNPACK, screen off - is consistent with the under-3-percent-per-hour target, but the figure has not been measured. Record one call-hour measurement during the build (R7) and put the screenshot in the deck; an unmeasured battery claim is a gift to a sharp judge.

7. Demo-Day Failure Modes and Fallbacks

For 60 seconds on stage, the demo is the product. Every failure below has a rehearsed fallback, and the final 6 hours of the schedule exist for exactly these rehearsals - not for features. The general rule: nothing on stage is live-dependency unless it has a tested plan B, and no clip appears that has not survived rehearsal on the actual device, in the actual room.

Table 7-1: Failure modes and rehearsed fallbacks

#	Failure	Chance	Rehearsed fallback
F1	Venue has no usable cellular signal for the live-call segment.	Medium	The FileSource scan of bundled audio runs the identical pipeline on stage; present it as the offline-proof segment rather than an apology (W7).
F2	The judge-voice clone cannot be generated in time.	Medium	Judge recording done 30 minutes before with the render queue ready; a rehearsal recording of a teammate serves as the backup clone subject.
F3	A cloned sample scores GREEN on stage.	Low after calibration	Only pre-tested clips and calibrated thresholds appear on stage; any clip that flakes in rehearsal is cut from the script (R3, C7).
F4	A teammate's genuine call scores RED.	Low to medium	Thresholds are asymmetric by design (R4); the exact teammate, device and room are tested the night before.
F5	The overlay is hidden by the OEM in-call UI.	Medium	Overlay z-order verified on the demo device in advance; fallback: bring MAYA to the foreground the moment the call ends, with the verdict replayed from the log.
F6	The OEM kills the service mid-call.	Low	Battery-optimization exemption granted during setup; service type verified on the demo device (W5).
F7	A judge asks about voice that is already public.	High (the question, not the failure)	Scripted answer: nothing retroactive exists anywhere; Shield starves future harvesting; Guard catches what slips through - defense in depth, no magic (C1).

8. Known Unknowns: Verify Before Hour 6

The research lookup tool was unavailable during the review, so the following facts are carried as assumptions rather than knowledge. Each has a verification method; owners are assigned at the sign-off meeting before hour 0. The rule is simple: after hour 6, no VERIFY item may remain load-bearing. Anything still unverified by then must be designed around, not hoped past.

Table 8-1: Facts to verify, why they matter, and how

#	Unknown	Why it matters	How to verify
U1	AASIST-L pretrained checkpoint availability and its ONNX-export status.	Determines the hour-0 conversion path and whether R1's fallback clock starts immediately.	Locate the checkpoint (official repository or a mirror), run the export dry-run, record parameter count and FP16 size.
U2	AntiFake-class perturbation model size and mobile feasibility.	Scopes every Shield claim on the roadmap (C6).	Inspect the open-source implementation; benchmark one clip on a mid-range phone and record the processing time.
U3	Vosk small and whisper-tiny int8 exact RAM footprint and Hindi/English accuracy.	Decides the scam-engine rung (R6, D4) and the RAM verdict in Table 6-1.	On-device benchmark with the target scripts in both languages; record RAM and word error rate.
U4	Adversarial perturbation survival through an AMR/Opus encode-decode roundtrip.	Calibrates the honest strength of every Shield claim (C6).	Encode-decode test: perturbed clip versus re-encoded perturbed clip; measure the clone-quality delta between them.
U5	TelecomManager.endCall() behavior on the demo OEM.	Determines whether the intervention ladder keeps step 3 (W3).	Device test with and without the dialer role; record which path actually hangs up.
U6	Actual call-audio access behavior on the demo device's Android version.	Underpins the capture design (W1, R2) - the entire product rests on this.	Run the capture-source matrix in the first 6 hours on the exact demo device.
U7	NNAPI or XNNPACK delegate availability and inference latency on the demo device.	Fixes the acceleration choice (W4) and the latency figures in Table 6-2.	Benchmark both delegates on the fixed demo device; lock XNNPACK as the default unless the GPU wins decisively.

9. Open Decisions Requiring Sign-Off

Six decisions block design freeze. Each carries the review's recommendation; if the sign-off meeting produces no objection, the recommendation stands. Decisions D1, D3 and D5 shape hour 0 itself, so they close first; D4 and D6 have explicit fallback clocks inside the schedule.

Table 9-1: Decisions, options, recommendations, deadlines

ID	Decision	Options	Recommendation	Deadline
D1	Call-audio capture strategy	(a) Speakerphone plus mic capture with auto-speaker; (b) chase downlink capture on one specific test device.	(a). Auto-enable the speaker on connect and show an explicit cannot-scan state for earpiece-only calls. Option (b) burns build hours against a platform wall (W1).	Before hour 6
D2	Privacy promise versus blocklist freshness	(a) Keep no-INTERNET, blocklist rides app releases; (b) restricted HTTPS update channel and the promise becomes audio-never-leaves.	(a) for the MVP: the absolute no-INTERNET invariant is the stronger judge story. Revisit (b) after the hackathon.	Before hour 0
D3	App shell technology	(a) Native Kotlin with Compose; (b) Flutter as drafted in the PRD.	(a): saves 20-40 MB against the RAM budget; the entire UI is one pill and one screen.	Day 0
D4	Scam-intent engine depth	(a) Keyword-spotting model only; (b) streaming ASR plus rules on the transcript; (c) no scam engine in the MVP.	Ladder: keyword spotting first, ASR if RAM and hours allow (R6); (c) only if the schedule collapses by hour 24.	By hour 12
D5	Verdict UX model	(a) Single RED/GREEN as in PRD v0.1; (b) staged amber, provisional, confirmed with an asymmetric RED.	(b): it matches the latency math (C3) and protects against false accusation (R4).	Day 0
D6	Model strategy	(a) Quantize the strong pretrained AASIST-L (FP16, then int8 with a representative calibration set); (b) start from a smaller model.	(a). Option (b) is the hour-12 export fallback (R1), not the default. Quantization without a representative set is forbidden - it silently costs accuracy.	By hour 12

10. Appendix A: Corrected Claims - PRD v0.1 versus Engineering Reality

Load-bearing sentences from PRD v0.1, the engineering reality behind each, and the revised wording proposed for v0.2. Items marked as surviving are confirmed correct and should be

kept verbatim; the appendix deliberately includes them so the team knows what does not need defending. Everything else should be edited before the PRD is shown outside the team.

Table 10-1: Line-by-line claim corrections

PRD v0.1 says	Engineering reality	Revised wording for v0.2
Real-time detection: first result under 4 seconds after the call connects.	The full-confidence vote completes at roughly 9-11 seconds; a 4-second result is single-window and provisional (C3).	Provisional verdict in about 4 seconds; confirmed verdict in about 9 seconds.
Always-on: no need to put on speaker or tap manually.	Downlink capture is blocked (W1); acoustic capture requires the speaker path.	Zero-tap protection: MAYA auto-enables the speaker on answer and scans; earpiece-only calls show an explicit cannot-scan state.
Audio capture: AudioSource.VOICE_COMMUNICATION at 16 kHz mono.	That source applies acoustic echo cancellation, which suppresses the far-end voice during speakerphone capture (R2).	16 kHz mono capture; the source is chosen empirically from MIC, UNPROCESSED and VOICE_COMMUNICATION per the hour-6 matrix test.
AASIST-Lite, TFLite Float16, about 22 MB.	Model weights are about 0.8-3 MB; the 22 MB figure only makes sense as a whole-pipeline budget.	AASIST-class model in FP16/int8: weights under 3 MB, whole-pipeline budget about 22 MB.
Verdict CLONED: full red takeover, HANG UP.	Programmatic hang-up requires dialer status (W3).	Full red takeover with a one-tap END CALL action and call-back-on-saved-number guidance.
Accuracy: 9 out of 10 cloned samples flagged correctly, live on stage.	Claimable only on the calibrated corpus (C7).	9 out of 10 on our channel-simulated calibration corpus; the live demo uses pre-tested material.
100 percent offline inference; zero internet permission.	Survives - OS-enforced and the strongest claim in the PRD.	Keep verbatim; lead with it in the pitch.
Privacy Vault: no audio saved, buffer wiped after inference.	Survives; extend with the scam-engine transcript note once D4 lands.	Keep, and add: scam analysis runs on-device; transcripts never leave the phone.
Explainable: tell WHY it is fake (no breath, flat pitch, spectral artifact).	Survives via DSP features; these three also survive telephony band-limiting.	Keep as is.
Hacker-proof by architecture: no DB, no server, nothing to leak.	Survives.	Keep verbatim.

Reading order for the team: Section 3 and Section 9 before hour 0; Section 5 and Section 8 assigned at the sign-off meeting; Section 7 rehearsed in the final 6 hours. Appendix A becomes the editing checklist for PRD v0.2.